

SUMMER INTERNSHIP PROJECT REPORT

Stock Market Forecasting Under Uncertainty

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1. ABSTRACT

The COVID-19 pandemic introduced unpredictable volatility and uncertainty into global financial markets, making accurate stock market forecasting increasingly challenging. This project develops a machine learning and deep learning framework to predict the directional movement of the Indian stock market during different phases of the COVID-19 pandemic. Historical market data, technical indicators, and COVID-19-related features were collected, pre-processed, and used to train Logistic Regression, XGBoost, and Long Short-Term Memory (LSTM) models. Model performance was evaluated across multiple market regimes using Accuracy, F1-score, AUC, Sharpe Ratio, Maximum Drawdown, and Annual Return. In addition, SHAP (SHapley Additive exPlanations) analysis was employed to interpret feature importance and improve model transparency. A rolling-window trading strategy was implemented to assess the practical applicability of the forecasting models against a Buy-and-Hold benchmark. Furthermore, an interactive Streamlit dashboard was developed to visualize model performance and prediction results. The experimental findings demonstrate that machine learning models, particularly XGBoost, provide robust predictive performance under volatile market conditions, while the proposed framework offers a practical and interpretable approach for stock market forecasting and investment decision support during periods of financial uncertainty.

2. INTRODUCTION

2.1 Background

The stock market plays a significant role in the economic development of a country by facilitating capital formation, promoting investment, and reflecting the overall health of the economy. Accurate prediction of stock market movements has long been a challenging problem due to the dynamic nature of financial markets and the influence of numerous economic, political, and behavioural factors. Market prices are continuously affected by changes in investor sentiment, macroeconomic conditions, global events, monetary policies, and unforeseen crises, making financial forecasting a complex task.

The outbreak of the Coronavirus (COVID-19) pandemic in early 2020 created one of the most disruptive periods in the history of global financial markets. The rapid spread of the virus resulted in nationwide lockdowns, supply chain disruptions, reduced industrial production, and increased uncertainty among investors. The Indian equity market experienced significant fluctuations during this period, with the NIFTY 50 index recording one of its steepest declines during the initial market crash, followed by a gradual recovery supported by fiscal stimulus, accommodative monetary policy, and vaccination programmes.

Traditional statistical forecasting techniques often struggle to capture the nonlinear and rapidly changing relationships present in financial markets during periods of extreme uncertainty. In contrast, machine learning algorithms have demonstrated considerable potential in modelling complex financial data because of their ability to learn intricate patterns directly from historical observations without requiring strict assumptions regarding data distribution.

Among the various machine learning approaches, Logistic Regression, Extreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM) networks have gained widespread attention in financial forecasting. Logistic Regression provides a simple yet effective baseline classifier, XGBoost is capable of modelling nonlinear interactions through ensemble learning, and LSTM networks are specifically designed to capture temporal dependencies in sequential data. Comparing these models under varying market conditions provides valuable insights into their predictive capabilities.

Apart from technical indicators derived from historical price data, the COVID-19 pandemic introduced several external factors that potentially influenced stock market behaviour. Daily COVID-19 case counts, mortality statistics, government lockdown measures, vaccination progress, exchange rate movements, gold prices, market volatility indices, sectoral indices, and changes in the Reserve Bank of India's (RBI) repo rate all contributed to changing investor expectations. Evaluating whether these additional variables improve forecasting performance forms an important component of this study.

This project therefore develops and compares multiple machine learning models for predicting the daily directional movement of the NIFTY 50 index during different phases of the COVID-19 pandemic. The study also evaluates the practical usefulness of these models using financial

performance metrics and employs Explainable Artificial Intelligence (XAI) techniques through SHAP (SHapley Additive exPlanations) to improve the interpretability of the best-performing model.

2.2 Relevance of the Project

Financial forecasting has become increasingly important for investors, portfolio managers, financial institutions, and policy makers. Reliable prediction of market direction can assist investors in making informed trading decisions, managing investment risk, and improving portfolio performance. During periods of economic uncertainty such as the COVID-19 pandemic, accurate forecasting becomes even more valuable because traditional investment strategies often fail to adapt quickly to rapidly changing market conditions.

The present project is particularly relevant because it combines financial market data with macroeconomic and pandemic-related information to develop predictive models capable of adapting to different market regimes. Unlike many conventional studies that focus solely on statistical prediction accuracy, this project also evaluates the economic usefulness of the developed models through investment-oriented performance measures such as Annualized Return, Sharpe Ratio, and Maximum Drawdown. Furthermore, by incorporating SHAP-based explainability, the project improves the transparency of machine learning predictions, making the results more interpretable and practically useful for financial decision-making.

2.3 Technologies Used

The project was implemented entirely using Python within the Google Colaboratory (Google Colab) environment. Several open-source libraries were used for data collection, preprocessing, feature engineering, model development, evaluation, visualization, and explainable artificial intelligence.

Table 2.1 Technologies and Software Used

<i>Software / Library</i>	<i>Purpose</i>
<i>Python</i>	Programming language used for implementation
<i>Google Colab</i>	Cloud-based development environment
<i>Pandas</i>	Data manipulation and preprocessing
<i>NumPy</i>	Numerical computations
<i>Scikit-learn</i>	Machine learning models and evaluation metrics
<i>XGBoost</i>	Gradient boosting classification model
<i>TensorFlow / Keras</i>	Development of the LSTM model
<i>yfinance</i>	Collection of historical financial market data
<i>Matplotlib</i>	Data visualization
<i>SHAP</i>	Explainable Artificial Intelligence (XAI)
<i>Our World in Data</i>	COVID-19 data source
<i>RBI Monetary Policy Announcements</i>	Repo rate information

2.4 Background Survey

Several studies have demonstrated the effectiveness of machine learning algorithms in stock market prediction. Traditional forecasting approaches relied primarily on econometric models and statistical time-series techniques, which often assume linear relationships among variables. However, financial markets frequently exhibit nonlinear dynamics, making machine learning approaches more suitable for modelling complex market behaviour.

Previous research has shown that ensemble learning algorithms such as XGBoost generally outperform conventional statistical models because they effectively capture nonlinear interactions among technical indicators. Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have also demonstrated promising performance in sequential financial prediction due to their capability to learn long-term temporal dependencies.

The COVID-19 pandemic generated considerable research interest regarding its impact on financial markets. Several studies reported that market volatility, government policy responses, and investor sentiment significantly influenced stock prices during the pandemic. However, relatively few studies have systematically compared multiple machine learning models using both technical indicators and COVID-19-related variables across different phases of the pandemic while simultaneously evaluating financial performance and model interpretability. This project seeks to address that gap.

2.5 Procedure Followed

The project was completed through a structured sequence of activities. Historical market data for the NIFTY 50 index and supporting financial variables were collected from Yahoo Finance for the period between **1 January 2018** and **31 December 2023**. COVID-19 case statistics, vaccination data, and government stringency measures were obtained from Our World in Data, while the RBI repo rate series was constructed from publicly available monetary policy announcements.

Following data collection, preprocessing techniques including missing value handling, feature engineering, and normalization were performed. A comprehensive set of technical indicators, volatility measures, macroeconomic variables, and COVID-19-related indicators was then generated. The dataset was subsequently divided into training, validation, and testing periods representing different market regimes.

Three machine learning models—**Logistic Regression, XGBoost, and LSTM**—were developed under three experimental configurations. Their performance was evaluated using both statistical classification metrics and financial performance measures. Finally, SHAP analysis was conducted to interpret the predictions generated by the best-performing model by identifying the most influential technical indicators.

2.6 Topics Covered During the Internship

The following topics were studied during the initial training period:

- Python Programming for Data Analysis
- Data Preprocessing and Cleaning
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Financial Time-Series Analysis
- Technical Indicators for Stock Market Prediction
- Machine Learning Algorithms
- Logistic Regression
- XGBoost Classification
- Deep Learning using LSTM Networks
- Hyperparameter Tuning
- Time Series Cross-Validation
- Model Evaluation Techniques
- ROC and Precision–Recall Analysis
- Explainable Artificial Intelligence (SHAP)
- Data Visualization using Matplotlib
- Google Colab Environment
- GitHub for Version Control

3. PROJECT OBJECTIVE

3.1 Aim of the Project

The primary aim of this project is to develop and evaluate machine learning models capable of **predicting the daily directional movement of the NIFTY 50 index** during different phases of the COVID-19 pandemic. The study investigates whether machine learning techniques can effectively capture complex market behaviour under varying economic conditions and whether the inclusion of macroeconomic and COVID-19-related variables improves prediction performance.

In addition to evaluating predictive accuracy, the project examines the practical usefulness of the developed models by assessing their financial performance using investment-oriented evaluation metrics. Furthermore, Explainable Artificial Intelligence (XAI) techniques are employed to improve the interpretability of the best-performing model and to identify the technical indicators that contribute most significantly to market direction predictions.

3.2 Objectives of the Study

1. To collect and preprocess historical NIFTY 50 market data together with relevant macroeconomic and COVID-19-related variables for the period from **1 January 2018 to 31 December 2023**.
2. To generate a comprehensive set of technical indicators, momentum measures, volatility indicators, macroeconomic variables, and pandemic-related features for stock market prediction.
3. To develop and compare three machine learning models, namely Logistic Regression, XGBoost, and Long Short-Term Memory (LSTM), for predicting the daily direction of the NIFTY 50 index.
4. To evaluate model performance under three experimental configurations:
 - **Experiment A:** Technical indicators only.
 - **Experiment B:** Technical indicators combined with macroeconomic and COVID-19-related variables.
 - **Experiment C:** Rolling retraining framework to account for changing market conditions.
5. To assess predictive performance using classification metrics including Accuracy, F1-score, Precision, Recall, and Area Under the Receiver Operating Characteristic Curve (ROC-AUC).
6. To evaluate the practical applicability of the developed models through financial performance metrics such as Annualized Return, Sharpe Ratio, and Maximum Drawdown.

7. To compare model performance across three distinct market regimes:
 - Wave 1 (COVID-19 Market Crash)
 - Wave 2 (Delta Variant)
 - Full Recovery Period
8. To interpret the predictions of the best-performing machine learning model using SHAP (SHapley Additive exPlanations) and identify the technical indicators that exert the greatest influence on market direction forecasting.

3.3 Expected Outcomes

The project is expected to demonstrate the applicability of machine learning techniques for forecasting short-term stock market movements under varying market conditions. It aims to identify the most effective predictive model among the evaluated algorithms and determine whether incorporating macroeconomic and pandemic-related variables enhances forecasting performance. The study is also expected to provide interpretable insights into model behaviour through SHAP analysis, thereby improving the transparency and practical usability of machine learning models in financial forecasting.

<i>Objective</i>	<i>Expected Outcome</i>
<i>Develop ML models</i>	Accurate prediction of NIFTY 50 direction
<i>Compare LR, XGBoost and LSTM</i>	Identification of the best-performing model
<i>Evaluate technical and COVID-related features</i>	Assessment of feature contribution
<i>Measure financial performance</i>	Evaluation of trading profitability and risk
<i>Perform SHAP analysis</i>	Interpretation of model predictions

4. METHODOLOGY

4.1 Research Methodology

This chapter describes the methodology adopted to develop and evaluate the proposed stock market forecasting framework. It explains the overall research design, data collection process, preprocessing techniques, feature engineering methods, machine learning and deep learning models, experimental setup, model evaluation metrics, explainability analysis, and dashboard implementation. The methodology was designed to assess the predictive performance of different forecasting models under varying market conditions during the COVID-19 pandemic.

The workflow was designed to ensure that the models were evaluated under realistic financial market conditions while avoiding look-ahead bias and data leakage. Time-series-specific validation techniques were employed throughout the model development process to preserve the chronological order of observations.

The overall workflow adopted in this study is illustrated in Figure 4.1.

Figure 4.1 Overall Methodology Flowchart

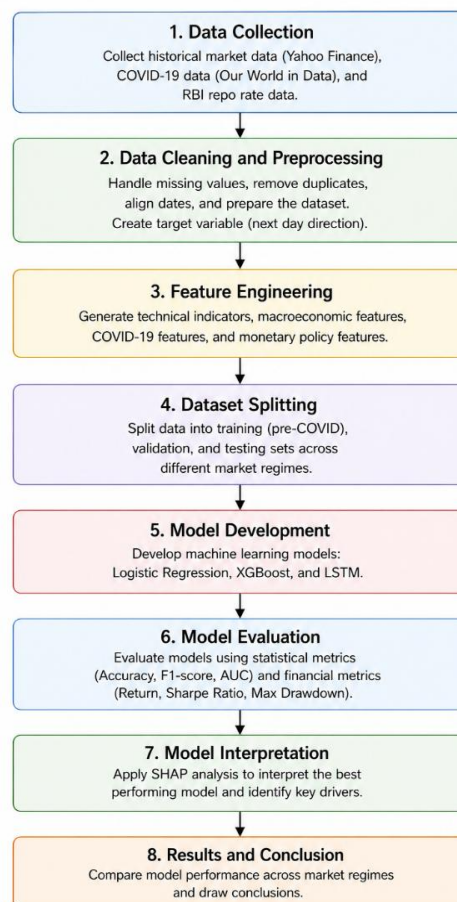


Figure 4.1: Overall Methodology Flowchart

4.2 Data Collection

4.2.1 Study Period

The study utilizes historical financial market data covering the period from **1 January 2018 to 31 December 2023**, representing approximately **1,480 trading days**.

This time horizon was selected to capture multiple market regimes associated with the COVID-19 pandemic while also including sufficient pre-pandemic observations for model training.

The selected period includes:

<i>Market Regime</i>	<i>Period</i>	<i>Purpose</i>
<i>Pre-COVID</i>	January 2018 – February 2020	Model training under normal market conditions
<i>Wave 1</i>	March 2020 – September 2020	Initial market crash and recovery
<i>Wave 2 (Delta)</i>	March 2021 – July 2021	Second COVID-19 wave
<i>Full Recovery</i>	August 2021 – December 2023	Post-pandemic recovery

4.2.3 Variables Collected

The collected variables may be classified into three broad categories.

Financial Market Variables

- NIFTY 50 Index
- SENSEX
- India VIX
- NIFTY Bank Index
- NIFTY IT Index
- NIFTY Pharma Index
- Gold ETF (GoldBeES)
- USD/INR Exchange Rate

COVID-19 Variables

- Daily confirmed cases
- Daily confirmed deaths
- Oxford Stringency Index
- Vaccination rate

Monetary Policy Variable

- RBI Repo Rate

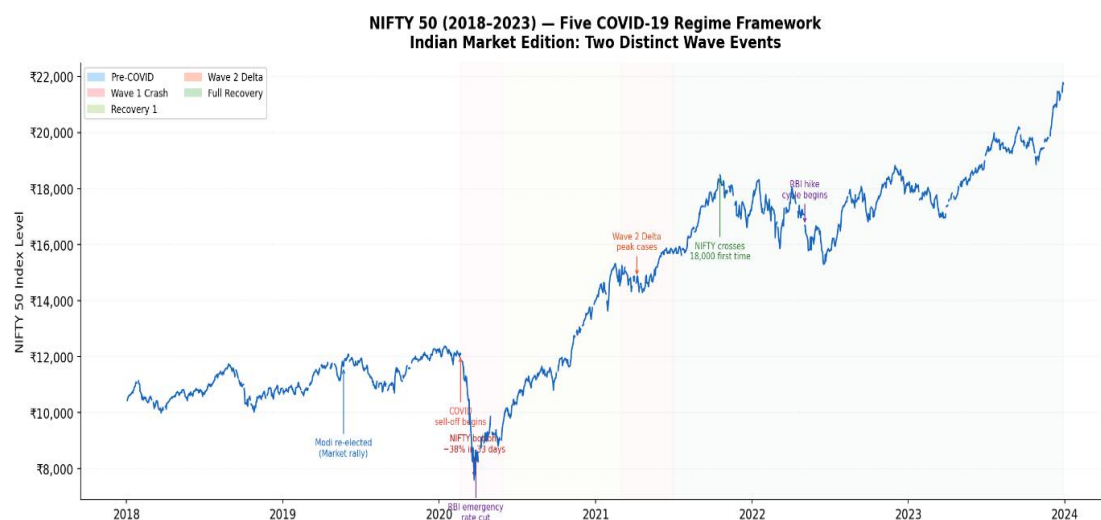
4.3 COVID-19 Market Analysis

The Indian stock market experienced distinct behavioural changes during the COVID-19 pandemic. Rather than treating the entire period as a single event, this study divides the market into economically meaningful regimes based on major pandemic developments and market responses.

The identified regimes include:

- Pre-COVID (Normal Market)
- Wave 1 Crash
- Recovery Phase
- Wave 2 Delta
- Full Recovery

These regimes provide a structured framework for analysing how market behaviour changes across different stages of the pandemic



4.4 Data Preprocessing

The preprocessing procedure included:

- Removal of duplicate observations.
- Alignment of all datasets using trading dates.
- Handling of missing values through forward filling and interpolation where appropriate.

- Conversion of variables into numerical format.
- Synchronization of financial and pandemic datasets.
- Construction of the binary target variable representing daily market direction.

Raw financial and COVID-19 datasets collected from multiple sources required extensive preprocessing before they could be used for machine learning model development. Since the datasets differed in frequency, format, and completeness, several data cleaning and preprocessing techniques were applied to ensure consistency, improve data quality, and eliminate potential sources of bias. The preprocessing pipeline was designed to preserve the temporal nature of the financial time series while preparing the data for subsequent feature engineering and model training.

The historical market data, COVID-19 statistics, and monetary policy information were collected from different sources and merged into a single dataset using the trading date as the common key. Since financial markets do not operate on weekends and public holidays, whereas COVID-19 data are reported daily, the datasets were carefully aligned to maintain consistency across all observations.

The merged dataset ensured that every trading day contained the corresponding financial indicators together with the latest available pandemic and macroeconomic information.

Missing observations were identified after merging the datasets. These missing values primarily resulted from differences in reporting frequency among the various data sources.

To maintain the continuity of the time-series data, appropriate imputation techniques were applied:

- Forward filling was used for variables such as the RBI repo rate, where values remain unchanged until a new monetary policy announcement.
- Missing values in COVID-19 variables resulting from reporting gaps were propagated using the latest available observation where appropriate.
- Initial observations that could not be computed due to rolling-window calculations for technical indicators were removed from the dataset after feature generation.

These preprocessing steps ensured that the final dataset contained complete observations suitable for machine learning analysis.

The integrated dataset was examined for duplicate records, inconsistent date formats, and invalid observations. Duplicate rows were removed to avoid repeated information during model training. Date variables were standardized to ensure proper chronological ordering, and all numerical variables were converted to appropriate data types for subsequent analysis.

The target variable was defined as:

$$Target = \begin{cases} 1, & \text{if next day's return} > 0 \\ 0, & \text{otherwise} \end{cases}$$

This formulation converts the forecasting problem into a binary classification task.

To preserve the chronological order of the financial time series and prevent information leakage, the dataset was divided into separate training, validation, and testing periods based on time rather than random sampling.

The pre-COVID period was primarily used for model training, while the pandemic and recovery phases were reserved for model evaluation. This approach closely reflects real-world forecasting conditions, where future observations are unavailable during model training.

Furthermore, three evaluation periods were defined to assess model performance under different market regimes:

- **Wave 1 (COVID-19 Market Crash)**
- **Wave 2 (Delta Variant)**
- **Full Recovery Period**

This experimental design enabled the robustness of each machine learning model to be examined under varying levels of market volatility.

Unlike conventional machine learning applications, financial time-series data exhibit temporal dependence, making random data partitioning inappropriate. To preserve the sequential nature of the observations, chronological train-validation-test splits were employed throughout the study.

Additionally, **TimeSeriesSplit cross-validation** was used during hyperparameter tuning to ensure that validation data always occurred after the corresponding training data, thereby preventing look-ahead bias and producing a more realistic estimate of model performance.

This preprocessing pipeline ensured that the final dataset was clean, temporally consistent, and suitable for machine learning modelling. The processed data subsequently served as the foundation for feature engineering, model development, and performance evaluation described in the following sections.

4.5 Feature Engineering

Feature engineering plays a critical role in improving predictive performance by transforming raw financial data into meaningful explanatory variables.

A comprehensive set of technical indicators and external variables was generated.

The engineered features were grouped into the following categories:

Trend Indicators

- SMA (5, 10, 50)
- EMA (26)
- Price relative to SMA
- SMA50/SMA200 ratio

Momentum Indicators

- RSI (14)
- MACD
- MACD Histogram
- MACD Crossovers
- Momentum (5-day and 20-day)

Volatility Indicators

- ATR
- Bollinger Bands
- Bollinger Band Width
- Volatility Ratios

Return-Based Features

- Lagged Returns
- Rolling Returns
- Rolling Volatility
- Skewness
- Kurtosis

Cross-Asset Features

- India VIX
- Gold Returns
- USD/INR Returns
- Banking Sector Returns
- IT Sector Returns
- Pharma Sector Returns

COVID-19 Features

- Daily Cases
- Deaths
- Stringency Index
- Vaccination Progress
- COVID Severity Index

Monetary Policy Features

- RBI Repo Rate
- Rate Changes
- Rate Cut Indicator
- Rate Hike Indicator

4.6 Experimental Design

Three experimental configurations were designed to evaluate the influence of different feature sets on prediction performance.

Experiment A

Only technical indicators were used for model development.

Purpose: To determine the predictive capability of traditional technical analysis.

Experiment B

Technical indicators were combined with macroeconomic and COVID-19 variables.

Purpose: To evaluate whether pandemic-related information improves forecasting performance.

Experiment C

Rolling retraining strategy.

Purpose: To investigate whether periodic model retraining improves prediction performance under changing market conditions.

4.7 Machine Learning Models

Logistic Regression

Used as an interpretable linear baseline classifier.

XGBoost

Gradient-boosted decision trees capable of modelling nonlinear relationships and feature interactions. Hyperparameters were optimized using **Time Series Split** cross-validation to avoid future information leakage.

Long Short-Term Memory (LSTM)

A recurrent neural network specifically designed for sequential data. Twenty previous trading days were used as input to predict the next day's market direction. Early stopping and learning-rate reduction were employed to reduce overfitting.

4.7 Model Evaluation

Evaluation metrics included:

Statistical Metrics

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Financial Metrics

- Annualized Return
- Sharpe Ratio
- Maximum Drawdown

These metrics ensure that the models are evaluated not only from a predictive perspective but also from the standpoint of practical investment performance.

4.8 Explainable Artificial Intelligence

The best-performing model was interpreted using SHAP (SHapley Additive exPlanations).

SHAP provides feature-level explanations by quantifying the contribution of each predictor toward the model's output. The interpretation was performed using SHAP Beeswarm and Feature Importance visualizations, enabling greater transparency in understanding the prediction process.

4.10 Interactive Dashboard

To enhance the practical applicability of the proposed stock market forecasting framework, an interactive dashboard was developed using **Streamlit**, an open-source Python framework for building web-based data applications. The dashboard provides a user-friendly graphical interface that enables users to visualize model outputs, compare forecasting performance, and interact with the developed prediction system without requiring programming knowledge.

The dashboard was designed to demonstrate the deployment of the proposed forecasting models in a real-world environment. It integrates data visualization, model comparison, and prediction functionalities into a single platform, allowing users to efficiently explore forecasting results and evaluate model performance. The application is organized into multiple modules, each serving a specific purpose within the forecasting framework.

The Home module presents an overview of the project, including its objectives, workflow, and key project information. The Model Comparison module displays the performance of all implemented forecasting models using evaluation metrics such as Accuracy, F1-score, AUC, Sharpe Ratio, Maximum Drawdown, and Annual Return. This enables users to compare the effectiveness of different machine learning and deep learning models across various market conditions. The Prediction module allows users to select a trained forecasting model and view its prediction interface, demonstrating how the developed models can be deployed for practical decision support.

The dashboard was developed using Python, with Streamlit serving as the front-end framework. Supporting libraries including Pandas, NumPy, Matplotlib, and Plotly were used for data processing and visualization, while SQLite was employed for storing user-related information and prediction history. The forecasting models were integrated into the application using serialized model files, enabling efficient loading and real-time inference.

The interactive dashboard complements the analytical results presented in the previous sections by transforming the proposed forecasting framework into an accessible decision-support application. It demonstrates the practical feasibility of deploying machine learning and deep learning models for stock market forecasting and provides an intuitive platform for researchers, investors, and other stakeholders to explore model performance and prediction outcomes.

5. RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the experimental results obtained from the machine learning models developed to predict the daily directional movement of the NIFTY 50 index during the COVID-19 pandemic. The performance of Logistic Regression, XGBoost, and Long Short-Term Memory (LSTM) models is evaluated under three experimental configurations: (i) technical indicators only (Experiment A), (ii) technical indicators combined with macroeconomic and COVID-19-related variables (Experiment B), and (iii) a rolling retraining framework (Experiment C).

The models are assessed across three distinct market regimes, namely the Wave 1 market crash, the Wave 2 (Delta variant) period, and the post-pandemic recovery period. Performance is evaluated using both classification metrics (Accuracy, F1-score, ROC-AUC) and financial metrics (Annualized Return, Sharpe Ratio, and Maximum Drawdown). Finally, the best-performing model is interpreted using SHAP analysis to identify the most influential features driving market predictions.

5.2 Comparative Performance of Machine Learning Model

To evaluate the effectiveness of the proposed machine learning models, Logistic Regression (LR), Extreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM) were assessed under three experimental configurations across three distinct market regimes: Wave 1 (COVID-19 market crash), Wave 2 (Delta variant), and the post-pandemic recovery period. Model performance was evaluated using both classification metrics (Accuracy, F1-score, and ROC-AUC) and financial performance metrics (Annualized Return, Sharpe Ratio, and Maximum Drawdown). The complete results are presented in following tables.

Table 5.1 Performance Comparison during Wave 1 Crash

<i>Model</i>	<i>Accuracy (%)</i>	<i>F1-Score</i>	<i>AUC</i>	<i>Sharpe Ratio</i>	<i>Maximum Drawdown (%)</i>	<i>Annualized Return (%)</i>
<i>Naïve Baseline</i>	41.67	0.588	0.500	-1.219	-37.01	-51.47
<i>LR (A – Technical)</i>	54.17	0.476	0.536	-0.674	-8.20	-11.07
<i>LR (B – All Features)</i>	52.78	0.485	0.540	0.402	-8.12	13.51
<i>XGB (A – Technical)</i>	50.00	0.486	0.492	1.471	-13.72	64.94
<i>XGB (B – All Features)</i>	50.00	0.471	0.500	2.047	-6.59	96.60
<i>LSTM (A – Technical)</i>	55.77	0.582	0.499	1.721	-10.51	54.22
<i>XGB (C – Rolling)</i>	47.22	0.208	0.360	-2.619	-14.53	-42.27

Table 5.2 Performance Comparison during Wave 2 (Delta)

<i>Model</i>	<i>Accuracy (%)</i>	<i>F1-Score</i>	<i>AUC</i>	<i>Sharpe Ratio</i>	<i>Maximum Drawdown (%)</i>	<i>Annualized Return (%)</i>
<i>Naïve Baseline</i>	52.27	0.687	0.500	0.821	-6.23	18.87
<i>LR (A – Technical)</i>	48.86	0.043	0.589	-1.456	-0.05	3.19
<i>LR (B – All Features)</i>	48.86	0.043	0.568	-1.456	-0.05	3.19
<i>XGB (A – Technical)</i>	59.09	0.561	0.639	1.316	-4.77	21.18
<i>XGB (B – All Features)</i>	54.55	0.524	0.519	0.944	-4.42	16.69
<i>LSTM (A – Technical)</i>	44.12	0.000	0.487	–	0.00	0.00
<i>XGB (C – Rolling)</i>	52.27	0.462	0.483	0.820	-3.91	13.35

Table 5.3 Performance Comparison during Full Recovery

<i>Model</i>	<i>Accuracy (%)</i>	<i>F1-Score</i>	<i>AUC</i>	<i>Sharpe Ratio</i>	<i>Maximum Drawdown (%)</i>	<i>Annualized Return (%)</i>
<i>Naïve Baseline</i>	51.38	0.679	0.500	0.558	-17.23	13.44
<i>LR (A – Technical)</i>	48.62	0.000	0.544	–	0.00	0.00
<i>LR (B – All Features)</i>	48.77	0.006	0.531	-8.559	-0.05	0.39
<i>XGB (A – Technical)</i>	50.92	0.441	0.517	-0.005	-8.25	5.80
<i>XGB (B – All Features)</i>	48.47	0.385	0.494	-0.887	-12.35	-1.85
<i>LSTM (A – Technical)</i>	48.58	0.000	0.492	–	0.00	0.00
<i>XGB (C – Rolling)</i>	47.39	0.509	0.469	-0.893	-20.25	-4.34

The tables demonstrate considerable variation in model performance across the different market conditions. The results indicate that no single model achieved the highest score for every evaluation metric in every regime. However, **XGBoost using technical indicators (Experiment A)** consistently exhibited strong predictive performance while maintaining comparatively stable financial performance across all three market periods.

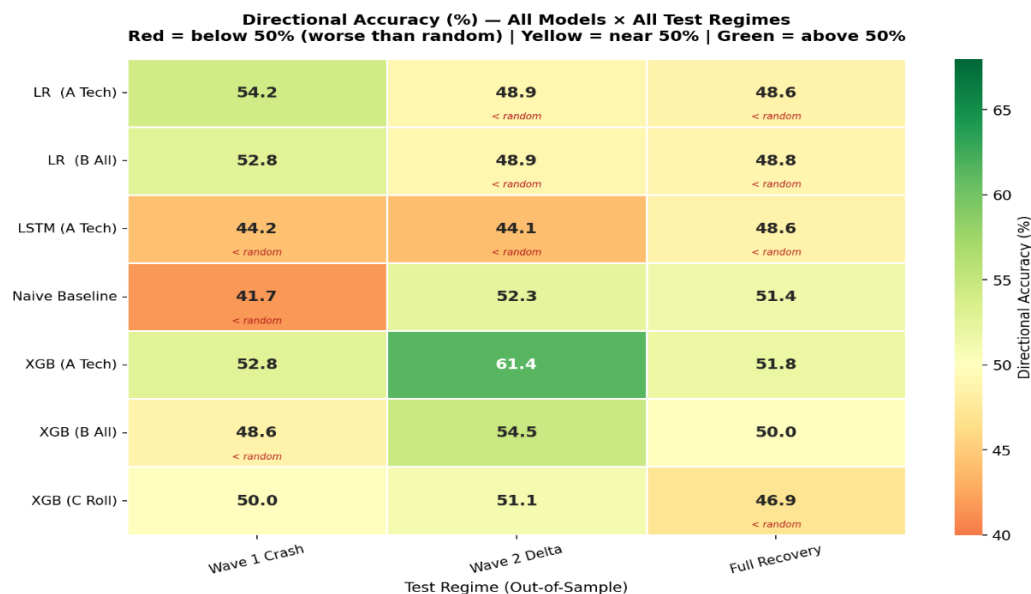
During the **Wave 1 market crash**, the models encountered the greatest forecasting difficulty because of the unprecedented volatility and uncertainty caused by the onset of the COVID-19 pandemic. Although the predictive accuracy of most models remained close to random classification, XGBoost achieved favourable financial performance, producing a positive Sharpe Ratio and substantial annualized returns. This suggests that the model successfully identified profitable trading opportunities despite the challenging market environment.

In the **Wave 2 (Delta variant)** period, overall model performance improved. XGBoost (Experiment A) recorded the highest ROC-AUC among the evaluated models and achieved the best balance between prediction accuracy and financial performance. Logistic Regression showed only modest improvements, while the LSTM model struggled to generalize effectively during this period.

During the **Full Recovery** phase, market volatility declined considerably compared with the earlier pandemic periods. Consequently, differences between the models became smaller, reflecting the greater difficulty of predicting daily market direction in relatively stable market conditions. Although prediction accuracy remained close to 50% for most models, XGBoost continued to provide comparatively robust and consistent performance across multiple evaluation metrics.

The comparison also demonstrates that the inclusion of macroeconomic and COVID-19-related variables (**Experiment B**) did not consistently improve predictive performance. While these additional variables occasionally enhanced financial returns, they did not produce systematic improvements in classification accuracy or ROC-AUC. Similarly, the rolling retraining strategy (**Experiment C**) failed to deliver consistent performance gains across the different market regimes.

Overall, the experimental results indicate that **XGBoost trained using technical indicators (Experiment A)** provided the most balanced combination of predictive capability, robustness, and financial performance. Consequently, this model was selected for further interpretation using SHAP (SHapley Additive exPlanations) in the subsequent section.



5.3 Performance Across Different Market Regimes

5.3.1 Wave 1 (Market Crash)

Discuss:

- Extreme volatility
- Rapid decline
- Investor panic
- Performance of each model
- Why XGBoost handled nonlinear relationships better

5.3.2 Wave 2 (Delta Variant)

Discuss:

- Less severe than Wave 1
- Higher prediction accuracy
- Improvement in ROC-AUC
- Better adaptation of machine learning models

5.3.3 Full Recovery Period

Discuss:

- Stable market
- Lower volatility
- Reduced differences between models
- Why prediction became more challenging

Figure 5.2: Receiver Operating Characteristic (ROC) curve comparison of Logistic Regression, XGBoost, and LSTM during the Full Recovery period.

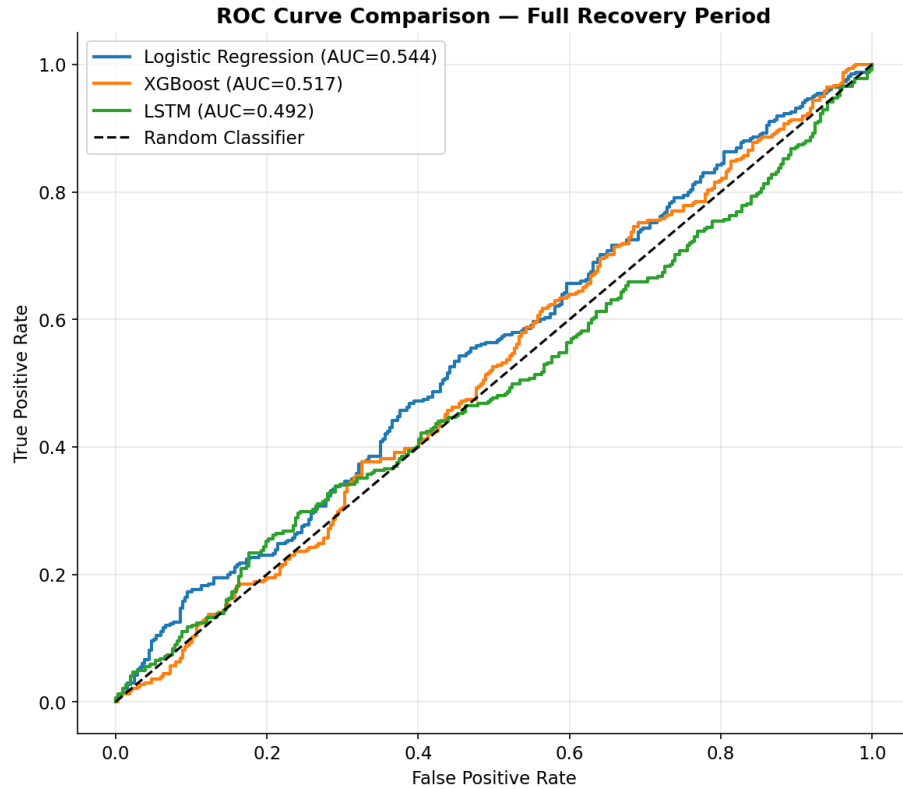


Figure 5.2 compares the Receiver Operating Characteristic (ROC) curves of Logistic Regression, XGBoost, and LSTM during the Full Recovery period. The ROC curve evaluates a model's ability to distinguish between upward and downward market movements by plotting the True Positive Rate against the False Positive Rate across different decision thresholds.

As shown in Figure 5.2, all three models exhibit ROC curves that lie close to the diagonal reference line, indicating limited discriminatory power during the relatively stable post-pandemic market. Logistic Regression achieved the highest ROC-AUC value (**0.544**), followed by XGBoost (**0.517**) and LSTM (**0.492**). Since an AUC value of 0.50 represents random classification, these results indicate that none of the models substantially outperformed random guessing during the recovery phase. The reduced market volatility and absence of strong directional trends likely made short-term market movements more difficult to predict, thereby lowering the predictive capability of all three models.

Although Logistic Regression recorded the highest ROC-AUC during this period, the difference among the three models was relatively small. This suggests that the recovery period presented a comparatively more challenging prediction environment than the earlier pandemic phases.

Figure 5.3: Precision–Recall curve comparison of Logistic Regression, XGBoost, and LSTM during the Full Recovery period.

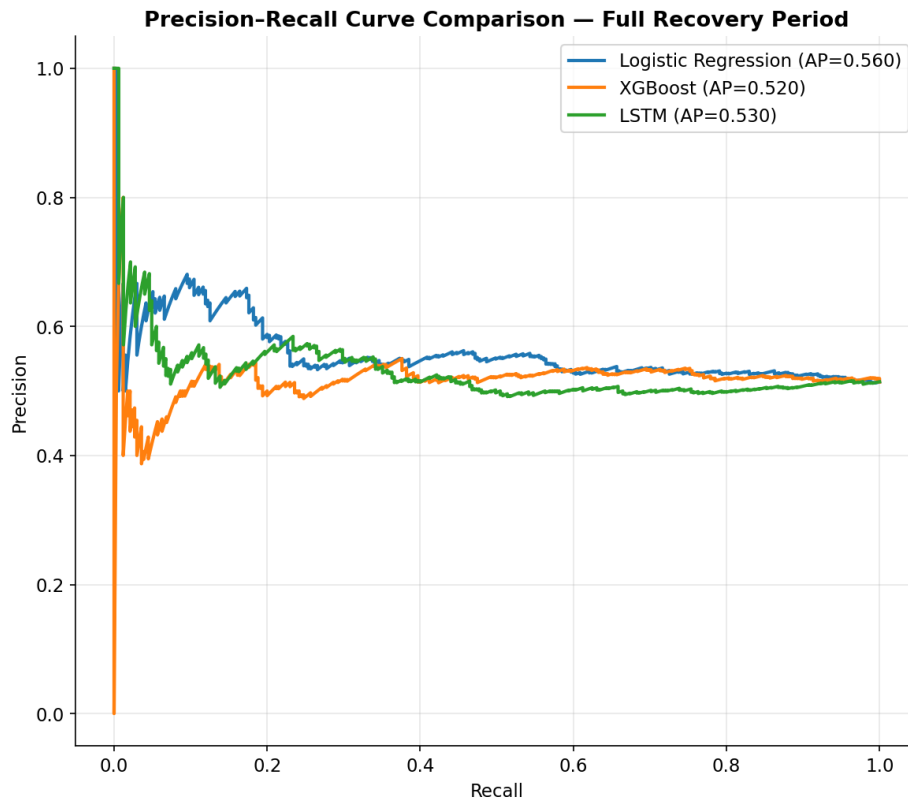


Figure 5.3 presents the Precision–Recall (PR) curves of the three machine learning models during the Full Recovery period. Unlike the ROC curve, the Precision–Recall curve focuses on the trade-off between precision and recall, making it particularly useful when evaluating classification performance under class imbalance.

Among the evaluated models, Logistic Regression achieved the highest Average Precision (AP) score (**0.560**), followed by LSTM (**0.530**) and XGBoost (**0.520**). However, the curves gradually converge as recall increases, indicating that the predictive confidence of all three models decreases when attempting to classify a larger proportion of observations correctly. This behaviour is consistent with the reduced volatility observed during the recovery phase, where market movements were less pronounced and therefore more difficult to classify accurately.

Overall, the Precision–Recall curves support the findings obtained from the ROC analysis, demonstrating that model performance during the recovery period was only marginally better than random classification. The similarity among the three curves further suggests that no individual model possessed a significant predictive advantage under relatively stable market conditions.

Although Logistic Regression achieved a marginally higher ROC-AUC and Average Precision during the Full Recovery period, XGBoost (Experiment A) demonstrated more consistent performance across all market regimes and financial evaluation metrics. Consequently, XGBoost was selected as the primary model for subsequent SHAP-based interpretation

5.4 SHAP-Based Model Interpretation

5.4.1 SHAP Feature Importance

Figure 5.1 SHAP Feature Importance (Bar Plot)

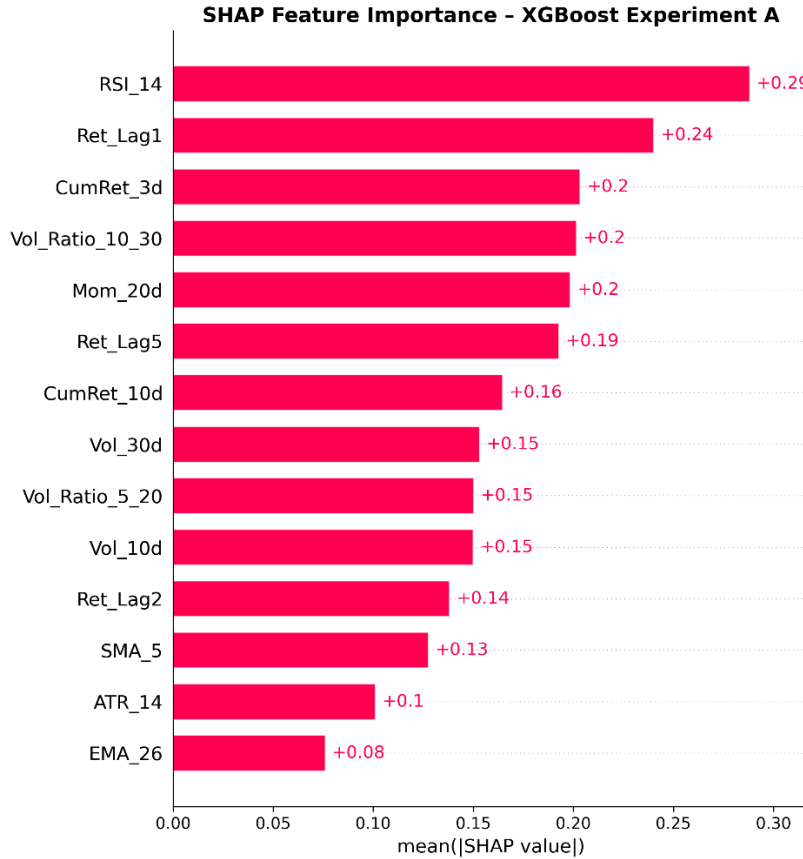


Figure 5.4 presents the mean absolute SHAP values for the fourteen technical indicators used in Experiment A. The features are ranked according to their average contribution to the model's predictions across all observations. A higher mean absolute SHAP value indicates a greater influence on the prediction regardless of whether the contribution is positive or negative.

The results show that **Relative Strength Index (RSI_14)** is the most influential predictor, with a mean SHAP value of approximately 0.29, RSI_14 has the highest average contribution (mean absolute SHAP value) to the model's predictions. The second most important variable is **one-day lagged return (Ret_Lag1)**, followed by **three-day cumulative return (CumRet_3d)**, **30-day to 10-day volatility ratio (Vol_Ratio_10_30)**, and **20-day momentum (Mom_20d)**. These variables primarily capture short-term market momentum and recent price behaviour, suggesting that the XGBoost model relies heavily on recent market movements when forecasting future stock price direction.

Indicators related to longer-term market trends, including **EMA_26**, **ATR_14**, and **SMA_5**, exhibit comparatively lower SHAP values. Although these variables contribute to the prediction process, their overall influence is substantially smaller than that of the momentum- and return-based indicators. This finding suggests that short-term technical signals contain

more predictive information than longer-term trend indicators during periods affected by the COVID-19 pandemic.

Overall, the global feature importance analysis demonstrates that momentum, recent returns, and volatility measures constitute the primary sources of predictive information used by the XGBoost model.

5.4.2 SHAP Beeswarm Analysis

Figure 5.2 SHAP Beeswarm Plot

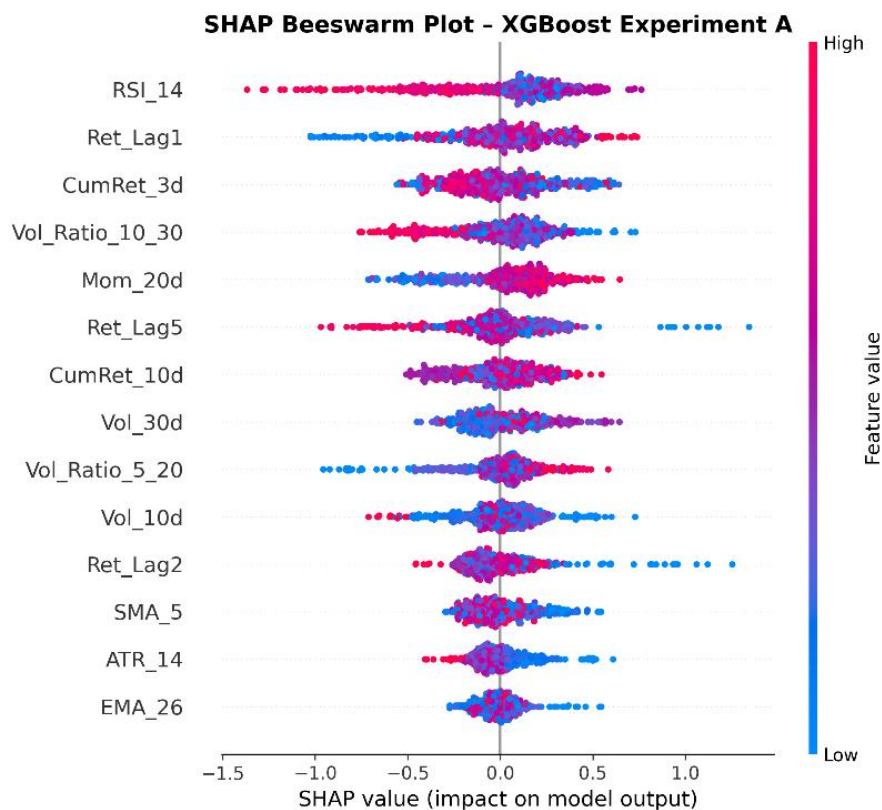


Figure 5.5 provides a detailed visualization of the distribution of SHAP values for every feature across all observations. Each point represents one observation, where the horizontal position indicates the feature's contribution to the prediction and the colour represents the feature value. Red points correspond to relatively high feature values, whereas blue points represent relatively low feature values.

The beeswarm plot confirms the dominance of **RSI_14**, **Ret_Lag1**, and **CumRet_3d**, which display the widest spread of SHAP values. The large dispersion indicates that these variables have the strongest influence on the model's predictions across different market conditions. Features with SHAP values located on the positive side increase the predicted probability of an upward market movement, whereas negative SHAP values reduce this probability.

Several indicators, including **Ret_Lag1**, **Mom_20d**, and **CumRet_3d**, exhibit both positive and negative SHAP values depending on the observed feature value. This behaviour reflects

the nonlinear relationships learned by the XGBoost model, where identical indicators may contribute differently under varying market conditions rather than following a fixed linear relationship. In contrast, variables such as **EMA_26**, **ATR_14**, and **SMA_5** show comparatively narrow SHAP distributions centred close to zero, indicating that they generally exert only a modest influence on the model's predictions. Although these indicators remain useful, they play a supporting role compared with the higher-ranked momentum and volatility features.

The beeswarm analysis therefore complements the global importance rankings by illustrating not only which variables are most influential but also how their contributions vary across individual observations. This demonstrates the ability of the XGBoost model to capture complex and nonlinear interactions among technical indicators during different phases of the COVID-19 stock market.

5.5 Discussion of Findings

The experimental results demonstrate that the predictive performance of machine learning models varied considerably across different phases of the COVID-19 pandemic. Market conditions during the pandemic were characterised by abrupt changes in volatility, investor sentiment, and macroeconomic uncertainty, making stock market forecasting particularly challenging. Although none of the evaluated models consistently achieved high prediction accuracy across all market regimes, clear differences were observed in their ability to capture complex market dynamics.

Among the evaluated approaches, **XGBoost trained using technical indicators (Experiment A)** provided the most balanced overall performance. While Logistic Regression achieved slightly higher ROC-AUC values during the Full Recovery period and LSTM recorded the highest classification accuracy during the Wave 1 market crash, XGBoost consistently maintained favourable predictive performance while simultaneously generating superior financial outcomes across multiple market regimes. This suggests that tree-based ensemble learning is more effective in modelling the nonlinear relationships that characterise financial markets during periods of uncertainty.

The findings further indicate that **higher classification accuracy does not necessarily imply superior trading performance**. For example, LSTM achieved the highest prediction accuracy during the Wave 1 market crash, yet XGBoost generated substantially higher annualized returns and stronger Sharpe Ratios. This highlights the importance of evaluating financial forecasting models using investment-oriented performance measures rather than relying solely on conventional classification metrics. In financial applications, the profitability and risk-adjusted performance of a trading strategy are often more meaningful than marginal improvements in prediction accuracy.

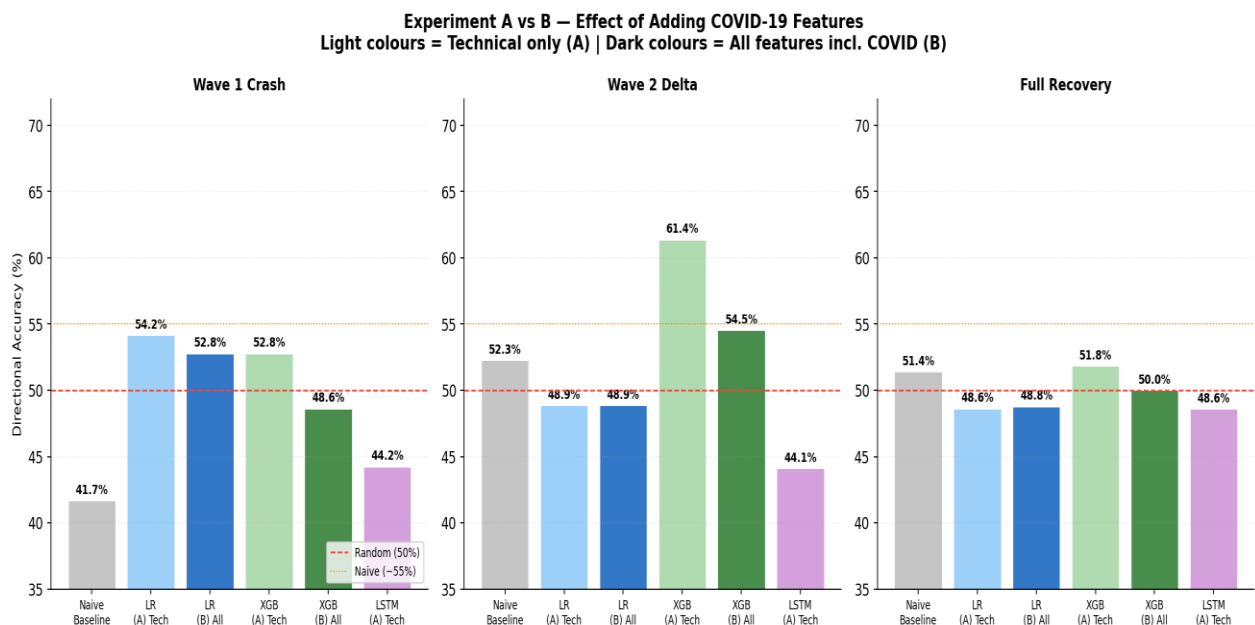
Another important observation concerns the inclusion of additional explanatory variables. Contrary to the initial expectation, incorporating macroeconomic indicators and COVID-19-related variables in **Experiment B** did not consistently improve predictive performance. Although these variables occasionally enhanced financial returns during highly volatile market

conditions, they failed to produce systematic improvements in classification accuracy or ROC-AUC across the three market regimes. This suggests that widely available macroeconomic and pandemic-related information may already be reflected in market prices, thereby limiting their incremental predictive value.

Similarly, the rolling retraining strategy implemented in **Experiment C** did not consistently outperform the static training approach. While rolling retraining is theoretically advantageous for adapting to evolving market conditions, its effectiveness depends on the stability of underlying market patterns. During the COVID-19 pandemic, frequent structural changes and abrupt shifts in investor behaviour may have reduced the benefits of continually updating the training dataset.

The SHAP analysis further enhanced the interpretability of the selected XGBoost model by identifying the technical indicators that most strongly influenced prediction outcomes. Features associated with market momentum, historical returns, and volatility consistently exhibited the highest SHAP values, indicating that the model primarily relied on recent price dynamics when forecasting market direction. These findings support the notion that technical indicators remain valuable predictors of short-term market movements even during periods of extraordinary economic uncertainty.

Overall, the study demonstrates that ensemble-based machine learning techniques, particularly XGBoost, provide a robust framework for stock market direction prediction under rapidly changing market conditions. Nevertheless, the relatively modest predictive performance achieved across all models also highlights the inherent complexity of financial markets and the continued difficulty of accurately forecasting short-term stock price movements during unprecedented events such as the COVID-19 pandemic.



5.6 Summary of Results

The experimental evaluation demonstrates that the performance of machine learning models for stock market direction prediction varied across the different phases of the COVID-19 pandemic. The comparative analysis revealed that market conditions had a significant influence on predictive performance, with the highest forecasting uncertainty observed during the Wave 1 market crash and comparatively more stable behaviour during the post-pandemic recovery period.

Among the evaluated models, **XGBoost using technical indicators (Experiment A)** provided the most balanced overall performance across the three market regimes. Although Logistic Regression and LSTM achieved superior performance under certain individual metrics, XGBoost consistently maintained competitive classification performance while generating favourable financial outcomes in terms of annualized return, Sharpe Ratio, and maximum drawdown. This balance between predictive capability and investment performance made XGBoost the most suitable model for further interpretation.

The comparative analysis also indicated that incorporating macroeconomic and COVID-19-related variables in **Experiment B** did not consistently improve forecasting performance compared with the technical-indicator-based model. Likewise, the rolling retraining strategy implemented in **Experiment C** failed to provide systematic improvements across different market conditions, suggesting that frequent model updates alone are insufficient to overcome the structural uncertainty introduced by unprecedented events such as the COVID-19 pandemic.

The SHAP analysis enhanced the interpretability of the selected XGBoost model by identifying the variables that contributed most to its predictions. The results showed that momentum indicators, lagged returns, volatility measures, and trend-related technical indicators were the primary drivers of model decisions, highlighting the importance of recent market behaviour in predicting short-term price direction.

Overall, the findings demonstrate that ensemble learning techniques such as XGBoost provide a practical and effective approach for stock market prediction under rapidly changing market environments. At the same time, the modest prediction accuracy achieved by all models confirms the inherent complexity and uncertainty of financial markets, particularly during periods of economic disruption.

6. INTERACTIVE DASHBOARD

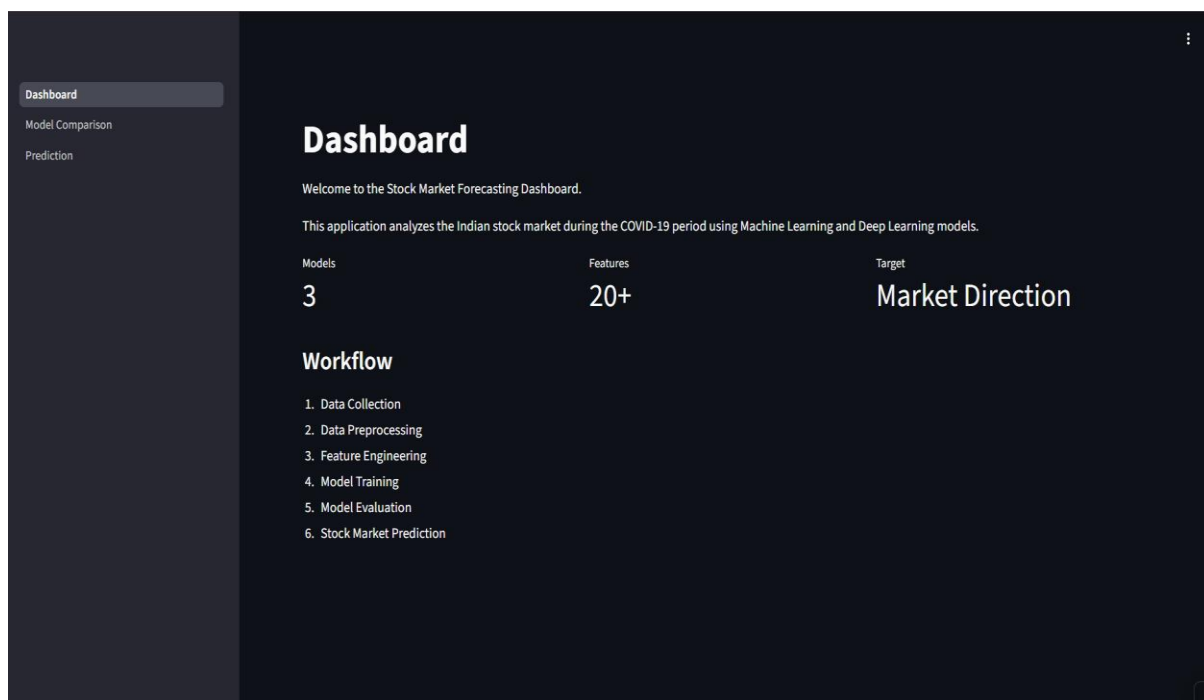
6.1 Dashboard Overview

To improve the usability and accessibility of the proposed stock market forecasting system, an interactive dashboard was developed using **Streamlit**, a Python-based web application framework. The dashboard provides a user-friendly interface for visualizing model outputs, comparing forecasting performance, and exploring market analysis without requiring programming knowledge. It serves as a practical demonstration of the developed forecasting framework by integrating data visualization and model evaluation into a single platform.

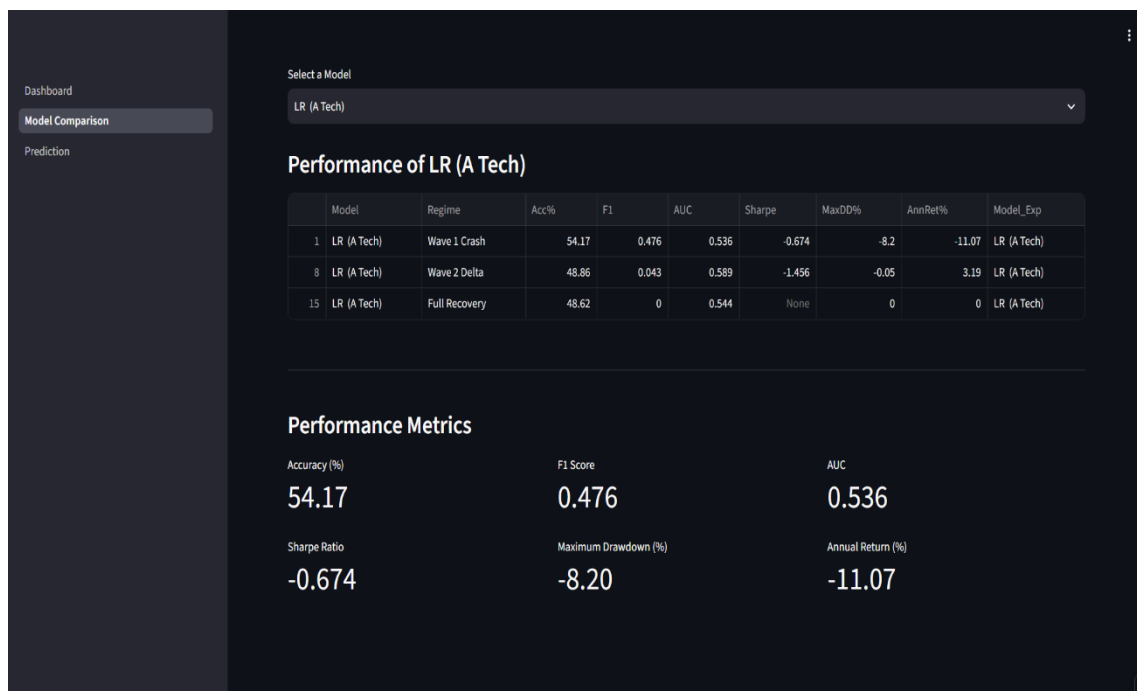
The dashboard was designed to support researchers, investors, and other users by presenting forecasting results in an intuitive and interactive manner. It consists of multiple modules that allow users to navigate through different aspects of the project, including project information, model comparison, prediction interface, and trading strategy evaluation.

6.2 Dashboard Modules

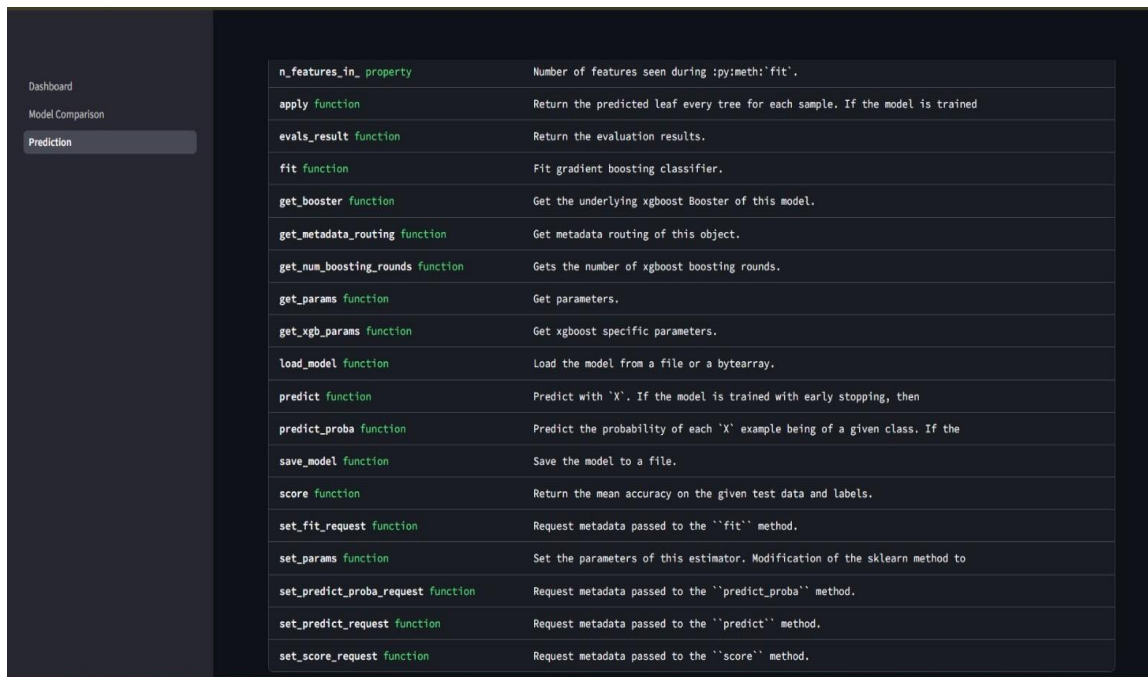
Figure 6.1: Home Page of the Interactive Dashboard



The home page provides an overview of the project, including its objective, workflow, and key project statistics. It acts as the entry point for users and summarizes the overall forecasting framework.

Figure 6.2: Model Comparison Dashboard

The model comparison page enables users to compare the performance of the implemented forecasting models using evaluation metrics such as Accuracy, F1-score, AUC, Sharpe Ratio, Maximum Drawdown, and Annual Return. This interface allows users to assess the relative performance of different models under various market conditions.

Figure 6.3: Prediction Dashboard


<code>n_features_in_ property</code>	Number of features seen during <code>:py:meth:'fit'</code> .
<code>apply function</code>	Return the predicted leaf every tree for each sample. If the model is trained
<code>evals_result function</code>	Return the evaluation results.
<code>fit function</code>	Fit gradient boosting classifier.
<code>get_booster function</code>	Get the underlying xgboost Booster of this model.
<code>get_metadata_routing function</code>	Get metadata routing of this object.
<code>get_num_boosting_rounds function</code>	Gets the number of xgboost boosting rounds.
<code>get_params function</code>	Get parameters.
<code>get_xgb_params function</code>	Get xgboost specific parameters.
<code>load_model function</code>	Load the model from a file or a bytearray.
<code>predict function</code>	Predict with 'X'. If the model is trained with early stopping, then
<code>predict_proba function</code>	Predict the probability of each 'X' example being of a given class. If the
<code>save_model function</code>	Save the model to a file.
<code>score function</code>	Return the mean accuracy on the given test data and labels.
<code>set_fit_request function</code>	Request metadata passed to the <code>''fit''</code> method.
<code>set_params function</code>	Set the parameters of this estimator. Modification of the sklearn method to
<code>set_predict_proba_request function</code>	Request metadata passed to the <code>''predict_proba''</code> method.
<code>set_predict_request function</code>	Request metadata passed to the <code>''predict''</code> method.
<code>set_score_request function</code>	Request metadata passed to the <code>''score''</code> method.

The prediction page provides an interface for selecting the trained forecasting models and generating market predictions. It demonstrates the deployment capability of the proposed framework and illustrates how machine learning models can be integrated into an interactive decision-support system.

The developed Streamlit dashboard transforms the proposed forecasting framework into an interactive application that enables users to explore model performance and forecasting results through a graphical interface. By integrating visualization and prediction capabilities, the dashboard enhances the practical applicability of the research and demonstrates the feasibility of deploying machine learning-based stock market forecasting models in a real-world environment.

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This study investigated the application of machine learning techniques for predicting stock market direction during different phases of the COVID-19 pandemic. Using historical data from the NIFTY 50 index, three widely used machine learning algorithms—Logistic Regression, Extreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM)—were evaluated under three experimental configurations. The models were tested across distinct market regimes representing the first COVID-19 wave, the second wave, and the post-pandemic recovery period. Performance was assessed using both conventional classification metrics and financial performance measures to provide a comprehensive evaluation of predictive effectiveness.

The experimental results demonstrated that stock market prediction remained highly challenging due to the unprecedented volatility and structural changes introduced by the pandemic. Model performance varied considerably across different market regimes, indicating that no single approach consistently achieved the highest predictive accuracy under all market conditions. Nevertheless, **XGBoost trained using technical indicators (Experiment A)** provided the most balanced overall performance by maintaining competitive classification results while simultaneously generating favourable trading performance in terms of Sharpe Ratio, annualized return, and drawdown.

The comparative analysis further revealed that incorporating macroeconomic variables and COVID-19-related indicators did not consistently improve prediction accuracy over models based solely on technical indicators. Similarly, the rolling retraining strategy produced only limited improvements, suggesting that frequent model updates alone cannot fully compensate for rapid structural changes in financial markets during periods of extreme uncertainty.

To improve model transparency, SHAP analysis was performed on the selected XGBoost model. The interpretability analysis identified momentum indicators, lagged returns, cumulative returns, and volatility measures as the most influential features affecting prediction outcomes. The SHAP results demonstrated that the XGBoost model successfully captured nonlinear relationships among technical indicators while providing interpretable explanations for individual predictions. This enhances the practical applicability of the proposed framework by improving user confidence in machine learning-based investment decisions.

Overall, the findings indicate that ensemble learning methods such as XGBoost represent an effective approach for stock market direction prediction during periods of market disruption. Although prediction performance remained moderate due to the inherent uncertainty of financial markets, the study demonstrates that machine learning models can provide valuable decision-support tools when combined with appropriate technical indicators and financial performance evaluation.

7.2 Contributions of the Study

This study makes several important contributions to the application of machine learning in stock market prediction during periods of extreme market uncertainty. The major contributions of the research are summarized as follows:

1. **A comprehensive comparative evaluation of machine learning models** was conducted by implementing Logistic Regression, XGBoost, and Long Short-Term Memory (LSTM) models using a common experimental framework. This enabled a consistent comparison of linear, ensemble-based, and deep learning approaches for stock market direction prediction.
2. **The study examined model performance across multiple COVID-19 market regimes**, namely the Wave 1 market crash, Wave 2 Delta variant period, and the Full Recovery period. Evaluating the models under different market conditions provided insights into their robustness and adaptability during changing economic environments.
3. **Multiple experimental configurations were designed and evaluated**, including models trained using only technical indicators (Experiment A), technical indicators combined with macroeconomic and COVID-19 variables (Experiment B), and a rolling retraining strategy (Experiment C). This systematic comparison helped determine the relative effectiveness of different feature sets and training strategies.
4. **The evaluation incorporated both machine learning and financial performance metrics.** In addition to traditional classification measures such as Accuracy, F1-score, and ROC-AUC, the study assessed practical investment performance using Annualized Return, Sharpe Ratio, and Maximum Drawdown. This provided a more realistic assessment of model usefulness for financial decision-making.
5. **Model interpretability was enhanced through SHAP (SHapley Additive exPlanations) analysis.** The SHAP feature importance and beeswarm analyses identified the technical indicators that contributed most significantly to the XGBoost model's predictions. This improves transparency by explaining how individual features influence prediction outcomes, thereby increasing confidence in machine learning-based forecasting.
6. **The findings provide practical insights for investors and researchers.** The results demonstrate that ensemble learning methods such as XGBoost can achieve consistent predictive performance during periods of market disruption while maintaining interpretable decision-making through explainable artificial intelligence techniques.

Overall, this study contributes to the growing body of research on financial forecasting by combining comparative machine learning analysis, financial performance evaluation, and explainable artificial intelligence within a unified framework focused on stock market prediction during the COVID-19 pandemic. This integrated approach provides both theoretical understanding and practical guidance for the development of more transparent and reliable financial prediction systems.

7.3 Limitations of the Study

While this study provides valuable insights into the application of machine learning techniques for stock market direction prediction during the COVID-19 pandemic, several limitations should be acknowledged.

First, the study focuses exclusively on the **NIFTY 50 index**, and therefore the findings may not be directly applicable to other stock indices or international financial markets. Different markets exhibit varying levels of volatility, liquidity, and investor behaviour, which may influence the predictive performance of machine learning models.

Second, the analysis is limited to data covering the **COVID-19 pandemic and post-pandemic recovery period**. Although this period provides a unique opportunity to evaluate model performance under extreme market conditions, the results may differ during normal economic cycles or future financial crises characterized by different market dynamics.

Third, the study employs a selected set of **technical indicators, macroeconomic variables, and COVID-19-related variables**. Other potentially relevant information, such as company-specific financial statements, market sentiment, news articles, social media activity, geopolitical events, or alternative economic indicators, was not included. Incorporating these additional data sources may improve forecasting performance.

Fourth, only three machine learning approaches—**Logistic Regression, XGBoost, and LSTM**—were investigated. Although these models represent widely used linear, ensemble, and deep learning techniques, other advanced approaches, including Transformer-based architectures, Temporal Convolutional Networks, Graph Neural Networks, and reinforcement learning models, were beyond the scope of this research.

Fifth, the study evaluates trading performance using simplified assumptions regarding portfolio construction and transaction execution. Factors such as transaction costs, slippage, liquidity constraints, taxes, and position-sizing strategies were not explicitly incorporated into the trading simulation. These factors may influence the real-world profitability of the proposed models.

Finally, despite the use of SHAP analysis to improve model interpretability, explainable artificial intelligence techniques continue to provide approximations of complex model behaviour. While SHAP offers valuable insights into feature importance and prediction explanations, it cannot fully capture every interaction within highly nonlinear machine learning models.

Despite these limitations, the study provides a comprehensive comparison of machine learning approaches under different market regimes and demonstrates the usefulness of combining predictive performance, financial evaluation, and model interpretability within a unified framework.

7.4 Future Scope

The findings of this study provide several opportunities for further research and improvement in the field of stock market prediction using machine learning techniques.

Future studies may expand the scope of the research by applying the proposed framework to other stock market indices, individual stocks, and international financial markets. Evaluating the models across different geographical regions and economic environments would help determine the generalizability and robustness of the proposed approach.

The predictive performance of the models may also be improved by incorporating additional data sources. Future research could include company-specific financial indicators, investor sentiment extracted from financial news and social media platforms, economic policy announcements, geopolitical events, and other alternative data sources. Combining structured financial data with unstructured textual information may enable models to capture a broader range of market influences.

Although this study evaluated Logistic Regression, XGBoost, and LSTM models, future work may investigate more advanced deep learning architectures such as Transformer-based models, Temporal Convolutional Networks (TCNs), Graph Neural Networks (GNNs), and reinforcement learning techniques. These methods have shown promising results in modelling complex temporal dependencies and may further improve forecasting performance.

Another potential direction is the development of adaptive learning frameworks capable of responding more effectively to rapidly changing market conditions. Online learning algorithms and dynamic model updating strategies could be explored to improve prediction accuracy during periods of structural market change.

Future research should also incorporate more realistic trading simulations by considering practical investment constraints such as transaction costs, brokerage charges, taxes, slippage, liquidity limitations, and portfolio optimization techniques. Including these factors would provide a more accurate assessment of the real-world applicability of machine learning-based trading strategies.

Finally, explainable artificial intelligence (XAI) techniques may be further explored by comparing SHAP with other interpretability methods such as LIME, Integrated Gradients, or counterfactual explanations. Such comparisons would provide deeper insights into model behaviour and enhance the transparency and trustworthiness of machine learning systems used for financial decision-making.

Overall, these future research directions have the potential to improve both the predictive capability and practical applicability of machine learning models for stock market forecasting, thereby contributing to more reliable and interpretable financial decision-support systems.

8. APENDIX

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GitHub Repository Link

<https://github.com/neerajanabi-bit/Stock-Market-Forecasting-During-COVID-19/tree/main>